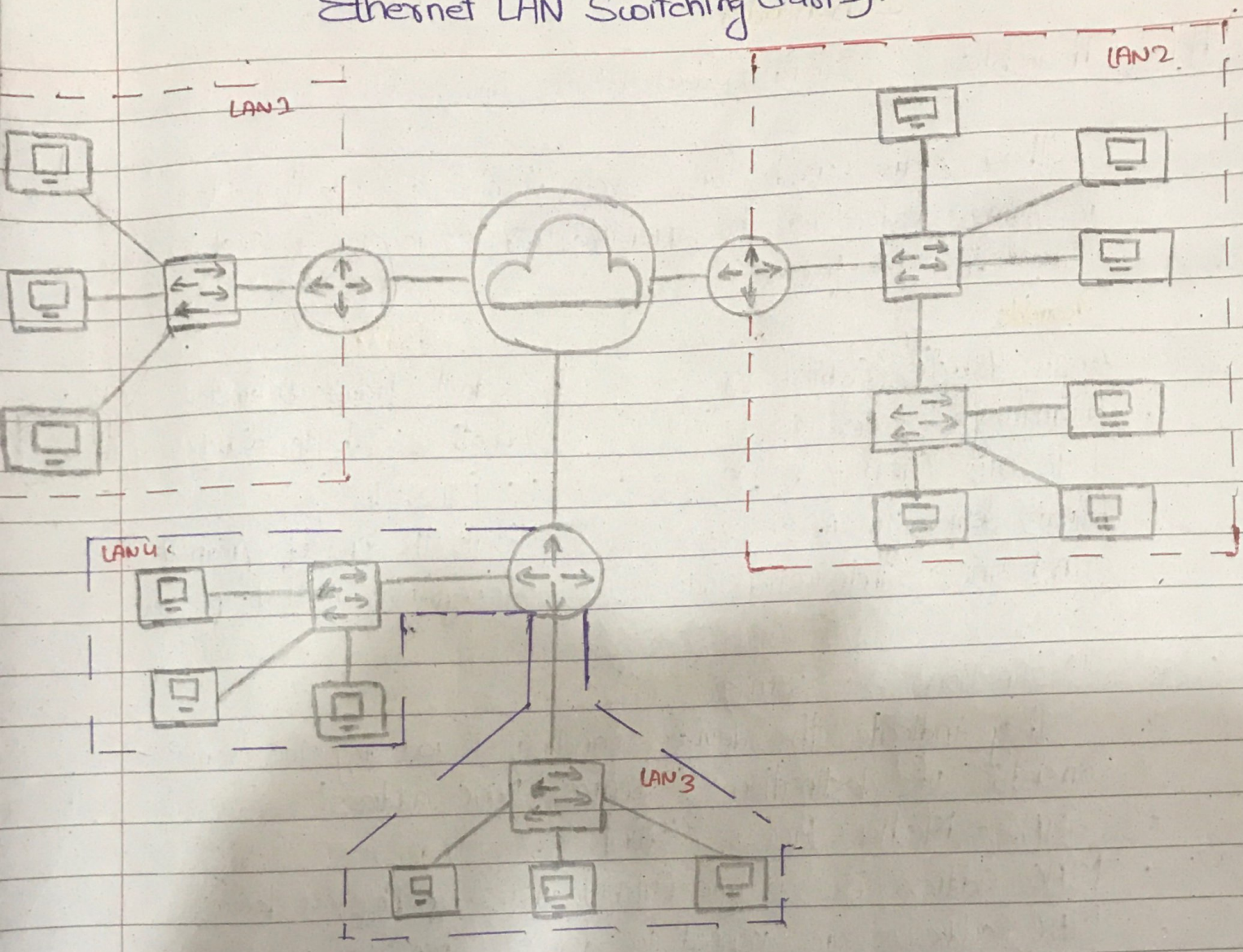
Day-5

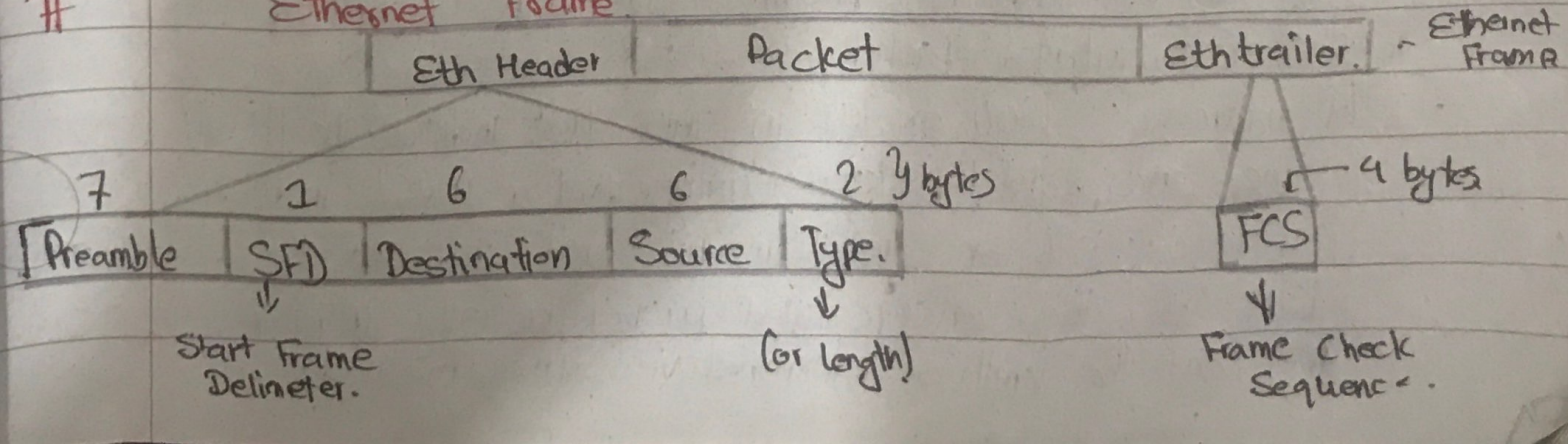
Oct-04

Ethernet LAN Switching (Part 1).



In LAN3 and Lan 4 2 switches are connected to different router interfaces, so they form 2 LAN. whereas in LAN 2, two switches are connected to each other, so they form a single LAN.

Ethernet Frame



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Eth. header.

Preamble & SFD (Start Frame Delimeter)

These are used for synchronization & to allow the receiving device to be prepared to receive rest of the data in frame.

Preamble

- Length: 7 bytes (56 bits)
- Alternating 1's and 0's
- 10101010 (7 बार)
- Allows devices to receive synchronize their receiver clock.

SFD

- Start Frame Delimeter.
- Length: 1 byte (8 bit)
- 10101011
- Marks the end of preamble & beginning of rest of frame.

Destination & Source

- They indicate the devices sending & receiving the frame.
- Consists of destination & source 'MAC address'.
- MAC - Media Access Control
- MAC address is 6 byte (48 bit) address of physical device
- MAC addresses are assigned to devices when they are made.

Type / Length.

- It is 2 byte / 16 bits in length.
- It can be used to represent either type or length of the encapsulated packet.
- A value of 1500 or less in this field indicates the LENGTH of the encapsulated packet (usually IPV4 or IPV6) (in bytes)
- A value of 1536 or greater in this field indicates the TYPE of the encapsulated packet (usually IPV4 or IPV6) and the length is determined via other methods.

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Eg. 1 For eg a value of 0x0800 which is written in hexadecimal & is equal to 2048 in decimal. The 0x infront of 0800 represent hexadecimal. 2048 > 1536, & it is used to identify that the encapsulated packet is an IPV4 packet.

Eg. 2 Similarly 0x86DD = 34525 in decimal is used to indicate the encapsulated packet is an IPV6 packet.

Eth. Trailer

Frame Check Sequence (FCS).

- Frame Check Sequence.
- 4 bytes (32 bits) in length.
- Detects corrupted data by running 'CRC' algorithm over the received data.
- CRC = 'Cyclic Redundancy Check'.

- Ethernet frame's Fram checker sequence is CRC.

MAC Address.

- 6-byte (48 bit) physical address assigned to the device when it is made.
- It is also known as 'Burned-IN-Address' (BIA).
- It is globally unique. (No two devices in the world should have the same MAC address).

Although there are MAC addresses known as 'locally-unique' MAC address, which don't have be gl to globally unique.

- The first 3 bytes are OUI (Organizationally Unique Identifier) which is assigned to company making device.
- The last 3 bytes are unique to the device itself.

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Dec	Hex
31	1F.
25	19
26	1A.

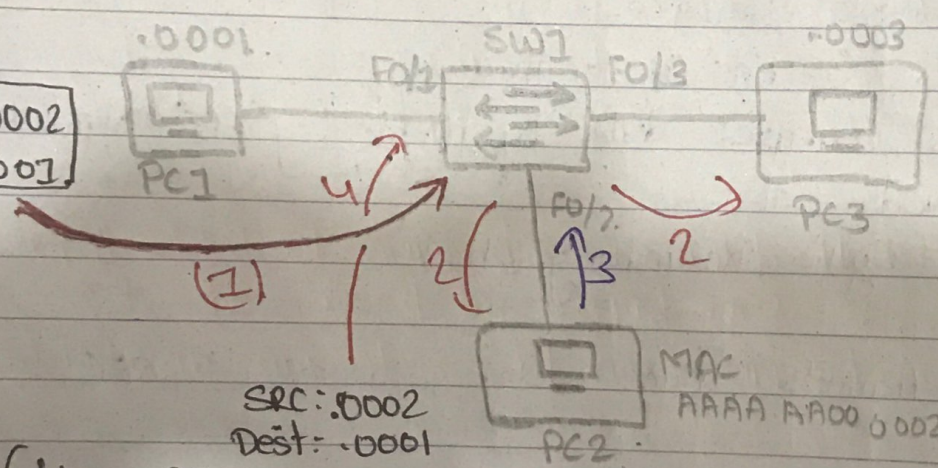
6. They are written as series of 12 hexadecimal characters. (6-hex char represent OUI & 6 are unique to device)

Hexadecimal												
0	1	2	3	4	5	6	7	8	9	A	B	C
D	E	F	10	11.	12	13	14	15	16	17	18	19

Decimal.												
0	1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24	25

Unicast Frame

Dest: .0002
Src: .0001



SRC: .0002
Dest: .0001

(Here AAAA.AA six char represent OUI)

Unicast Frame: a frame destined for single target.
In our case above, target is PC2.

MAC Address Table.		
	MAC	Interface.
1)	.0001	Fo/1
3)	.002	Fo/2.

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PC1 sends packet to SW1. SW1 adds the mac-address of PC1 on its MAC address table.

This is known as dynamically learned MAC address or Dynamic MAC address. because it wasn't manually configured on switch. The switch learned it itself.

Every switch keep a MAC address table and fill it dynamically by looking at SRC Maddress of frames it receives. Since, SW1 received a frame from SRC MAC .0001 it it fills the MAC address on table.

The destination MAC address of frame is .0002 but SW1 doesn't know where that is. This is called unknown unicast frame; a frame for which switch doesn't have entry in its MAC address table

Now, switch doesn't know how to reach destination. So, it FLOOD the frame (forward frame out of All its interfaces except the one from where it received packet, in our case PC1).

SW1 copies frame & send out of its F0/2 & F0/3 interface

Now, PC3 ignores the packet, because the destination MAC address doesn't match the its own MAC address, it simply drops packet.

PC2 however receives packet and processes it as per OSI stack. Unles PC2 sendsa a reply of some sort, it stops there. SW1 never receives a packet from PC2, so it can't learn PC2's MAC address

Now, if PC2 sends a frame out of its interface to dst .0001 and SW1 receives it, then it adds the address of PC2 in its table.

This is known as known unicast frame, as destination (.0001) is already in its MAC address table.

Unknown unicast frames are flooded whereas known unicast frames are simply forwarded to their destination.

In Cisco switches, dynamic MAC addresses are removed from table after 5 mins of inactivity

QUIZ

1) Which field of an ethernet frame provides receiver clock synchronisa
a) Preamble b) SFD c) Type d) FCS

2) How long is the physical address of a network device?
a) 32 bytes b) 32 bits c) 48 bytes d) 48 bits

3) What is the OUI of this MAC address? E8BA.7011.2874.
a) E8BA b) E8BA.70 c) 7011 d) E8BA.7011

4) Which field of an Ethernet frame does a switch use to populate its MAC address table?
a) Preamble b) length c) source MAC address d) Destination MAC addr

5) What kind of frame does a switch flood out of all interfaces except the one it was received on?
a) unknown unicast b) known unicast c) All cast.

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